

EECS 348- Assignment 1
GenAI Code Analysis
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GenAI Programs

GenAI #1: ChatGPT (OpenAI). I accessed ChatGPT website on safari.

GenAI #2: Google Gemini. I accessed this through Google.

Prompt

Write a C program that picks a fixed secret number between 1 and 10. The user gets up to 3 tries to guess it. Use printf to prompt the user and scanf to read the guess.

Use a while or for loop for up to 3 guesses. Use if-else statements to say whether the guess is too high, too low, or correct. Stop immediately when the user guesses correctly, and print a final win or lose message. The text should look like this:

Guess a number between 1 and 10.

Attempt 1/3. Enter your guess: 5

Too low! Try again.

Attempt 2/3. Enter your guess: 8

Too high! Try again.

Attempt 3/3. Enter your guess: 7

Correct! You win!

Generated Code from ChatGPT #1

```
#include <stdio.h>
```

```
int main() {  
    int secretNumber = 7;  
    int guess;  
    int attempts = 0;  
    int correct = 0;  
  
    printf("Guess a number between 1 and 10.\n");  
  
    while (attempts < 3) {  
        attempts++;  
  
        printf("Attempt %d/3. Enter your guess: ", attempts);  
        scanf("%d", &guess);  
  
        if (guess == secretNumber) {  
            printf("Correct! You win!\n");  
            correct = 1;  
            break;  
        }  
    }  
}
```

```

    }
    else if (guess < secretNumber) {
        printf("Too low! Try again.\n");
    }
    else {
        printf("Too high! Try again.\n");
    }
}

if (correct == 0) {
    printf("You lose! The secret number was %d.\n", secretNumber);
}

return 0;
}

```

Generated Code from Google Gemini

```
#include <stdio.h>
```

```

int main() {
    int secret = 7; // Fixed secret number
    int guess;
    int won = 0; // Flag to track if the user guessed correctly

    printf("Guess a number between 1 and 10.\n");

    for (int attempt = 1; attempt <= 3; attempt++) {
        printf("Attempt %d/3. Enter your guess: ", attempt);
        scanf("%d", &guess);

        if (guess == secret) {
            printf("Correct! You win!\n");
            won = 1;
            break; // Stop loop immediately on correct guess
        } else if (guess < secret) {
            if (attempt < 3) {
                printf("Too low! Try again.\n");
            } else {
                printf("Too low!\n");
            }
        } else {
            if (attempt < 3) {
                printf("Too high! Try again.\n");
            } else {

```

```

        printf("Too high!\n");
    }
}

if (!won) {
    printf("Game over! You ran out of attempts. The secret number was %d.\n", secret);
}

return 0;
}

```

Code Analysis

ChatGPT:

Correctness: The program correctly handles normal integer guesses from 1-10.

It correctly shows the guesses as too high, too low, or correct, and it stops when the user guesses correctly. It also gives a final lose message after three incorrect guesses.

It however does not handle invalid non-integer input with `scanf()`.

Time complexity: The program uses a while loop that runs at most 3 times.

So its time complexity is $O(1)$ because the number of attempts is fixed and does not get bigger based on the input size.

Space complexity: The program uses only a few integer variables like: `secretNumber`, `guess`, `attempts`, and `correct`. It doesn't use arrays, dynamic memory, or recursion.

So, its space complexity is $O(1)$.

Maintainability: The program has simple control flow and clear variable names. The while loop, if, else if, and else statements are easy to follow and modify.

It would be more maintainable though if the AI gave more detailed comments and input validation.

Gemini:

Correctness: The program correctly handles normal integer guesses from 1–10. It identifies guesses as too high,

too low, or correct, stops immediately when the user guesses correctly, and also gives a final game-over message after three incorrect guesses which are all good.

It also doesn't does not handle invalid non-integer inputs.

Time complexity: The program uses a for loop that runs at most 3 times. So its time complexity is $O(1)$ because the number of attempts is fixed.

The nested if statements do not change the overall complexity because they execute only a fixed number of times.

Space complexity: The program uses a fixed number of integer variables and does not use arrays, dynamic memory, or recursion.

So its space complexity is $O(1)$. It uses slightly fewer variables than the other program which was from ChatGPT, but this does not make a meaningful memory advantage.

Maintainability: The program has clear variable names and some comments that explain most parts of the code.

But it has more nested if statements, especially for the final attempt.

That makes the control flow more complicated and could make future changes harder if I were comparing it with the simpler ChatGPT program.

Selecting starting code:

I picked the ChatGPT program as the starting point for my final program.

I think that both programs meet the main requirements of the assignment and both do have $O(1)$ time and $O(1)$ space complexity.

The reason I wanted to use the ChatGPT program is its maintainability.

Its control flow is simpler because it uses one while loop and a straightforward if-else structure.

The Gemini program uses more nested if statements for the final attempt and that makes the logic a bit more complicated.

I think that the simpler structure of the ChatGPT program will make it easier to understand and for me to modify it.

Improvements

For correctness, I will improve the input handling by checking whether `scanf()` successfully reads an integer. If the input is invalid then the program will display an error message instead of just continuing without actually dealing with the input.

For time complexity, I don't think that a major change is necessary because the program already has $O(1)$ time complexity. The loop can only run three times so the number of operations will still be limited.

For space complexity also won't have major changes because the program already uses $O(1)$ space.

For maintainability, I will add the required program prologue and detailed comments explaining the code from the instructions. I will also keep the simpler control flow structure from the ChatGPT program instead of adding other nested conditions.